JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SAMAYAVALLI ARUMUGAM PET.BUNK

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

Route Summary:
Total Distance: 40.04 km
Estimated Duration: 0.8 hours

Adjusted Duration (Heavy Vehicle): 1.1 hours
Start: (11.4381, 77.8734)
End: (11.649059, 78.119294)

Welcome to the Journey Risk Management Study

Route Safety Analysis Report

1. Overview of the Route Map

The route from Sangari, Tamil Nadu to TNHB Colony, Kandhampatty, Salem, Tamil Nadu, spans approximately 40 km. The journey predominantly follows National Highway 544 (NH544), a major roadway facilitating easy access due to its connectivity and fewer intersections. This highway is a preferred route for heavy vehicles and transport of goods due to its direct path and relatively high road standards.

2. Typical Weather Conditions and Hazards

The region experiences a tropical climate, with temperatures peaking in summer months from March to June. Monsoon typically occurs from July to September, posing challenges like reduced visibility and slippery road conditions due to heavy rainfall. It is important to be prepared for sudden weather changes, particularly during the monsoon season, which can cause flash flooding and disrupt travel.

3. Traffic Patterns

Traffic congestion is generally observed during peak hours from 8:00 AM to 10:00 AM and 5:00 PM to 7:00 PM, particularly near urban areas and entries to Salem. Weekends may witness increased traffic due to leisure travel. NH544 tends to have moderate traffic but can become congested due to large volumes of goods transport vehicles.

4. Assessment of Road Quality and Infrastructure

NH544 is a well-maintained highway, with multiple lanes and proper signage. However, some stretches, especially near smaller towns, may experience potholes and uneven surfaces. Continuous monitoring during monsoon is essential as roads can degrade faster. Rest stops and fueling stations are adequately spaced for convenience and safety.

5. Suggestions for Alternative Routes

In case of emergencies, consider using State Highway 86 (SH86) which runs parallel to sections of NH544 and provides access to smaller towns. This secondary route can serve as a detour but may have narrower lanes and less frequent maintenance.

6. Summary of Local Regulations

Transporting hazardous materials requires adherence to Tamil Nadu's Motor Vehicles Rules, which align with national standards on transport safety. This includes proper storage, using prescribed routes, and avoiding residential zones when possible. Ensure compliance by checking if there are permits or special restrictions specific to materials being transported.

7. Historical Incidents

While this is a well-traveled route, incidents involving heavy vehicles are primarily due to driver error or weather-related factors. Ensure that drivers are trained in defensive driving and are aware of emergency protocols in case of spills or accidents.

8. Environmental Considerations

The region is dotted with agricultural areas and reserves, necessitating caution to prevent contamination. Be mindful of wildlife crossings, especially in less urbanized stretches. Avoid idling to limit pollution and consider the impact of noise in populated areas.

9. Communication Coverage

Overall, the route enjoys good mobile network coverage, but rural stretches, particularly in hilly areas, may experience signal drops. Equip vehicles with offline GPS navigation systems and maintain both direct and alternative means of communication, like CB radios.

10. Emergency Response Times

Emergency services are generally accessible within 20-30 minutes in urbanized sections closer to Salem. In rural sections, response times could extend to 45 minutes to an hour. Familiarize drivers with first-response procedures and equip them with emergency kits.

12. Overall Summary of Risk Assessment

The route from Sangari to Salem is generally safe for the transport of hazardous materials but requires careful planning and awareness of local conditions. Peak traffic, weather patterns, and emergency protocols should be taken into account to mitigate risks. Continuous communication and updated weather and traffic reports can aid in minimizing potential hazards.

This assessment aims to provide comprehensive guidance to ensure safety and compliance during transit. Proper adherence to the strategies outlined will optimize safety across this transit route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.45007, 77.87201	15 KM/Hr
5	Turn	High	11.47407, 77.86839	15 KM/Hr
6	Turn	Medium	11.48366, 77.88781	30 KM/Hr
7	Turn	High	11.49149, 77.89601	15 KM/Hr
8	Turn	High	11.49224, 77.89601	15 KM/Hr
9	Turn	Medium	11.60942, 78.10056	30 KM/Hr
10	Turn	Medium	11.61736, 78.10588	30 KM/Hr
11	Turn	Medium	11.61753, 78.10608	30 KM/Hr
12	Turn	Medium	11.64007, 78.11794	30 KM/Hr
13	Turn	Medium	11.64020, 78.11786	30 KM/Hr
14	Turn	Medium	11.64497, 78.12028	30 KM/Hr
15	Turn	High	11.64503, 78.12034	15 KM/Hr
16	Blind Spot	Blind Spot	11.64952, 78.12062	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
12	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
13	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
14	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
15	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
18	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
22	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
23	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
16	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
17	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
19	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
20	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
21	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45007, 77.87201

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47407, 77.86839

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.48366, 77.88781

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49149, 77.89601

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49224, 77.89601

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60942, 78.10056

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61736, 78.10588

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61753, 78.10608

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64007, 78.11794

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64020, 78.11786

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64497, 78.12028

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64503, 78.12034

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.64952, 78.12062